

DIGITAL GUITAR

Increase Accessibility of the Guitar for Individuals with Reduced Pinch Strength

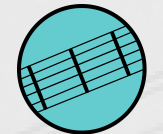
OBJECTIVES

↓ σ Minimize Fretting Strain

 Reduce Necessary Key Pinch Strength

 Play of Common Keys & Chords

 Be Authentic



Pressed Fret

+



Strummed String

Signal Mapping

Computer Algorithm

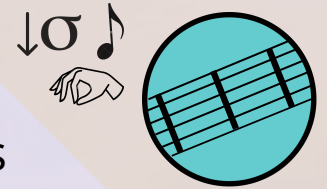


Musical Output
Speakers

Fretting Sensors

Tactile Sensors

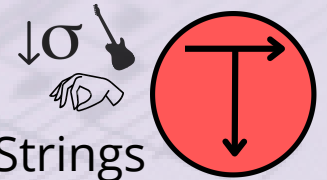
- Detects frets being held down
- Requires 92% less force to fret than regular guitar
- 1 Piezoresistive Tactile sensor per fret, creating a 6x12 grid



Low Tension Strings

Low Gauge Nylon Strings

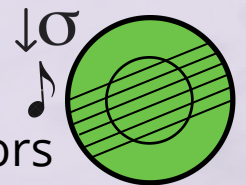
- Untuned strings to allow playing with minimal force



Strumming Sensors

Strain Gauge Sensors

- Detect tensile forces in strummed strings
- Magnitude of deformation is converted into electrical resistances for mapping output volume



Required Force to Fret Guitar Per String

